Tectonophysics, 148 (1988) 11–27
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Paleomagnetics of Ringerike Old Red Sandstone and related rocks, southern Norway: implications for pre-Carboniferous separation of Baltica and British terranes

D.N. DOUGLASS

Department of Earth Sciences, Dartmouth College, Hanover, NH 03755 (U.S.A.) *

(Received December 1, 1986; revised version accepted August 11, 1987)

Abstract

Douglass, D.N., 1988. Paleomagnetics of Ringerike Old Red Sandstone and related rocks, southern Norway: implications for pre-Carboniferous separation of Baltica and British terranes. *Tectonophysics*, 148:11–27.

Paleomagnetic study of two stratigraphic sections of the Upper Silurian–Lower Devonian Ringerike Sandstone, Oslo area, Norway, indicate a straightforward multicomponent NRM for these rocks. A lower temperature magnetization component gives a southwest and up (reversed polarity) in-situ direction, yields a negative fold test (i.e. post-folding) at the 99% confidence level, and is nearly identical to directions obtained from overlying Permo-Carboniferous lavas. A high temperature magnetization component exhibits both normal and reversed polarity directions, passes a fold test at the 95% confidence level, and is supported by a baked contact test which indicates a pre-intrusion age of magnetization. This characteristic high temperature component was successfully isolated in 79 samples, from 19 sites within the Ringerike succession. Results indicate a tilt corrected, characteristic magnetic direction of declination = 26°, inclination = −16°, which is consistent with earlier work. This direction corresponds to a time-integrated paleopole position of 19°N latitude, 164° longitude ($dp = 4.8°$, $dm = 9.3°$). When compared to previous results from rocks of similar age from the British Isles, an approximately 20° latitudinal displacement between Britain and Baltica during Late Silurian–Early Devonian time is suggested.

Introduction

Current geologic and paleomagnetic evidence suggests that large separations existed between North America and parts of Europe in the Early Paleozoic. By Late Devonian time however, the North American and European paleoplates were essentially assembled into the pre-Atlantic rift configuration of Bullard et al. (1965), representing the Euramerican part of the Pangea supercontinent (Kent and Keppie, in press). Late Silurian through Early Devonian time marks the youngest age in which large separations between constituent pieces of Euramerica could have still existed. Therefore, paleomagnetic evidence of this age should provide the best criteria for pre-Pangea plate geometry and subsequent plate assembly.

Paleomagnetic data from Late Silurian–Middle Devonian age rocks of Norway (Storetvedt et al., 1967; Storetvedt and Gjellestad, 1966; Lie et al., 1969), when compared to paleomagnetic directions from the British Isles (e.g. Briden et al., 1973, 1984; Tarling, 1985), suggest a large (~ 20°) latitudinal displacement between Baltica (represented by the Norwegian rocks) and the British terranes. The postulated tectonic boundary between these two regions lies beneath the North Sea, out of reach of close geologic scrutiny (Briden et al., 1973).

* Currently at the Department of Geology, Pasadena College, Pasadena, CA 91106 (U.S.A.).

0040-1951/88/$03.50 © 1988 Elsevier Science Publishers B.V.

There has been some degree of uncertainty attached to the Norwegian data for two principal reasons: (1) Previous Norwegian paleopole positions coincide with paleopole positions from Middle to Late Devonian age rocks from the British Isles (Briden et al., 1973). This suggested the possibility that all Norwegian Old Red Sandstone rocks were completely remagnetized after Early Devonian time (e.g. Creer, 1968). Recently, remagnetization has been demonstrated for the Old Red Sandstone of the Kvamshesten Basin (western Norway), which probably occurred during Late Devonian–Early Carboniferous time, concurrent with thrusting of the sequence (Torsvik et al., 1986). Although the Ringerike Sandstone occurs outside of the main Caledonide thrust belt, it is possible that these rocks have also been remagnetized. (2) No paleomagnetic field tests (such as fold tests, baked contact tests, etc.) exist for the Norwegian results to constrain the age of magnetization.

This investigation presents new paleomagnetic data for the Upper Silurian–Lower Devonian Ringerike Sandstone at its type locality near Oslo, Norway. These data are supported by paleomagnetic field tests which constrain the age of the magnetic components observed. Results are then compared to previous work from the British Isles in order to test the hypothesis of large-scale latitudinal offset between the British terranes and Baltica in Late Silurian–Early Devonian time.

Geologic setting

Siluro-Devonian sediments

The Ringerike Group is a laterally discontinuous sequence of Old Red Sandstone type clastic sediments exposed within the Oslo area of Norway (Fig. 1). In the type area (the area of the present study), Ringerike rocks conformably overlie the interbedded marine carbonates and shales of the Steinsfjorden Formation, which is of Late Wenlockian–Early Ludlovian age (Worsley et al., 1983). The base of the Ringerike Group is defined by the transition from drab colored marine rocks to reddish interbedded marine and non-marine lithologies (Turner, 1974a). Based on thelodont fauna, this lithologic transition is believed to approximate closely the Wenlock–Ludlow boundary (Turner and Turner, 1974). The general age aspect of the Ringerike fauna is considered Ludlovian; however, some comparison has been made to the Downtonian fauna of the Midland Valley of Scotland (Heintz, 1969; Turner and Turner, 1974).

The Ringerike Group has been divided into the Sundvollen Formation and Stubdal Formation (Fig. 2) at the type section (Turner, 1974a). The lower Sundvollen Formation consists of lithologies indicative of meandering stream and intermittent shallow marine deposition on a coastal alluvial plain (Turner, 1974a, b). The overlying Stubdal Formation consists of lithologies reflecting braided stream deposition, including both channel and overbank facies of a source-distal alluvial complex (Turner, 1974a).

Permo-Carboniferous lavas and sediments

The Ringerike Group and underlying Cambro-Silurian marine rocks are separated from the overlying Asker Group sediments by an angular unconformity (Figs. 1B and 2B). The Asker Group includes sediments representative of meandering and braided fluvial environments, shallow marine deposition, and locally, volcanic derived sandstones and braided stream conglomerates (Henningsmoen, 1978; Olaussen, 1981). The fresh-water flora and fauna of the Asker Group indicate an Early Permian age (Henningsmoen, 1978). However, a more recent study of foraminifera from a marine limestone within the Asker Group suggests a Middle Carboniferous age (Olaussen, 1981).

Conformably overlying the sediments of the Asker Group is the well known Permian lava sequence of the Oslo Graben. In the Ringerike area, the lowest extrusive igneous unit is an aphyric, quartz tholeiite basalt, (basalt-B1), which is one of the oldest lavas in the sequence (Larsen, 1978). A sample of this unit from the Krokskogen area has been dated by Rb/Sr whole rock analysis at 292 ± 8 Ma (Sundvoll, 1978). This indicates an uppermost Carboniferous age (Palmer, 1983) for the onset of extrusive igneous activity. The remainder of the sequence consists largely of rhomb porphyry flows with occasional interbedded basalt flows and sediments.

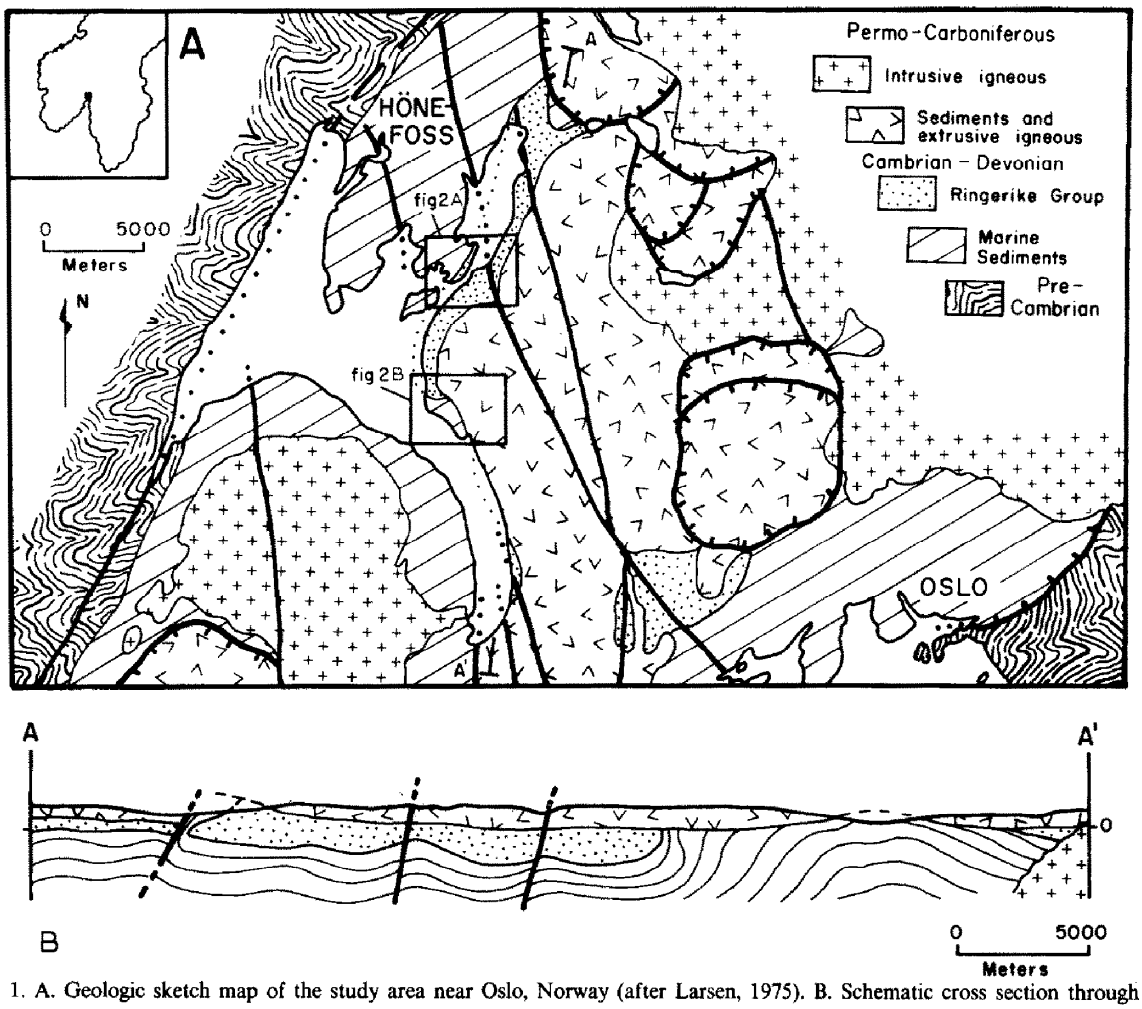


Fig. 1. A. Geologic sketch map of the study area near Oslo, Norway (after Larsen, 1975). B. Schematic cross section through the Ringerike area (after Holtedahl, 1953; Larsen, 1975; Morley, 1986).

Structure

Cambrian through Silurian sediments of the Oslo area were deposited in a foreland basin located southeast of the main Caledonian orogenic belt (Ramberg, 1976). The sequence was then folded, with the severity of deformation generally decreasing from northwest to southeast. There is a marked difference in structural style between the tight-to-isoclinally folded Cambro-Silurian marine carbonates and shales, and the open folding of the Ringerike Group sandstones and mudstones (Ramberg and Spjeldnaes, 1978). The structure of the pre-Carboniferous sedimentary succession within the Oslo graben has been recently interpreted as the buried thrust front of the lowest southern Norwegian Caledonide thrust sheet (Morley, 1986). However, significant transport by thrusting of rocks in the Ringerike area has yet to be demonstrated. The age of this phase of deformation cannot be younger than Late Carboniferous, and is thought to be Early Devonian based on correlation with other Caledonian structural events to the west (Ramberg and Spjeldnaes, 1978).

Permo-Carboniferous rocks show dominantly normal faulting associated with graben formation which began prior to the eruption of the first upper Carboniferous lavas (Ramberg and Spjeldnaes, 1978; Oftedahl, 1978). Broad open folding about east–west trending axes is also evident in the Carboniferous sediments and lavas (Larsen, 1975). Therefore, Ringerike rocks have undergone at least two stages of deformation, one demonstrably pre-Late Carboniferous, and the other concur-

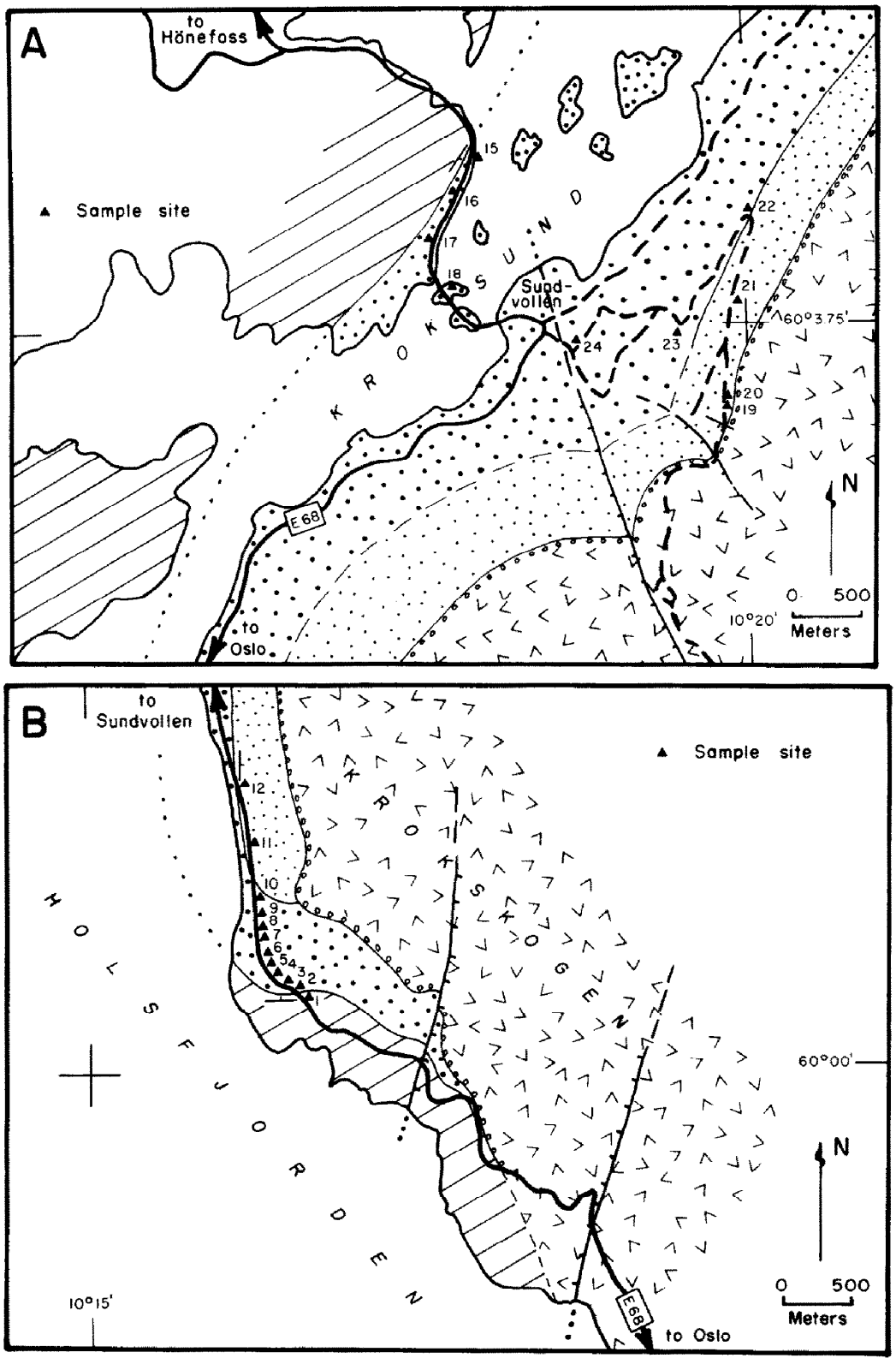


Fig. 2. Geologic location map of sample sites for (A) Section A, near the town of Sundvollen and (B) Section B, along Highway E-68, south of Sundvollen. Heavy stipple is the Sundvollen Formation, light stipple is the Stubdal Formation (after Turner, 1974a). Other symbols as in Fig. 1.

rent with and/or postdating Permo-Carboniferous sedimentation and volcanism.

Thermal history

Conodont alteration index values have been obtained for Ordovician and Silurian sediments in the Oslo area (Bergstrøm, 1980; Aldridge, 1984). Near the type section of the Ringerike Group, conodont alteration index values range from 3 to 4, indicating maximum paleotemperatures of between 110° and 300°C, with temperatures generally decreasing southward (Aldridge, 1984). Data also exist on the clay composition of numerous volcanic ash beds which occur in the upper part of the Silurian marine rocks, immediately below the base of the Ringerike. Hagemann (1966) reports only non-expandable illite and chlorite clay compositions from these bentonites. Furthermore, a bentonite sample collected from the Steinsfjord Formation in the northern part of the Ringerike area was found to consist of nearly pure non-expandable illite, with traces of chlorite (R.C. Reynolds, pers. commun., 1986). This assemblage indicates maximum temperatures of between 200° and 300°C (Hoffman and Hower, 1979), which is consistent with the conodont evidence. The maximum thickness of lavas and sediments ever to overly the Ringerike sequence is approximately 3 km (Oftedahl, 1978). Assuming a geothermal gradient of 25°C/km, this would indicate maximum burial related paleotemperatures of only about 75°C. Therefore the paleotemperatures observed are probably related to increased igneous activity during Permo-Carboniferous time.

Paleomagnetic methods

Sampling

Three stratigraphic sections of the Ringerike Group were selected and sampled for paleomagnetic study based on the mapping and stratigraphy of Turner (1974a). Results from two of these sections are reported here. Section A begins at the transition from marine to non-marine rocks on the west shore of Kroksund, near the town of Sundvollen (Fig. 2A). At this locality, bedding strikes northeast and dips 50° to 60° to the southeast, with bedding attitudes swallowing upsection. A sharp change in bedding attitudes between rocks on the western side of Kroksund (Fig. 2A), and outcrops on the islands in Kroksund suggests a possible fault. Sample sites were located at approximately equal stratigraphic intervals and were continued up section until Ringerike rocks become truncated by the overlying Carboniferous beds. It is likely that this section is broken by one or more post-Carboniferous faults having relatively small offsets.

Section B begins in excellent new exposures of the uppermost marine rocks along Highway E-68, approximately 8 km south of Sundvollen (Fig. 2B). At the base of the section, beds strike nearly east–west and dip steeply (60°–80°) to the north. However, bedding attitudes become shallower upsection, striking nearly north and dipping 20° to 25°E in the uppermost sites. Section B occurs on the south limb of a syncline which plunges 20° to approximately N60°E (derived from π diagram analysis of bedding attitudes). Carboniferous sediments which overlie the upper part of section A and all of section B (Fig. 2) strike N356°E and dip 20°E, Post-Carboniferous tilting of the sequence can account for the plunge observed in the Ringerike syncline at Section B.

Several dikes of probable Permo-Carboniferous age cut through the base of section A. These dikes clearly crosscut Ringerike bedding planes and are generally near vertical. In Section A (site 16), one of these dikes was sampled for a paleomagnetic baked contact test along with the surrounding Ringerike sediments. The Permo-Carboniferous lava sequence was sampled at three stratigraphic intervals; B-1, corresponding to the lowermost basalt; RP-7, a rhomb porphyry unit from the middle of the sequence and; RP-10, one of the upper lava units. Because Ringerike sediments were not found directly below any of the Permo-Carboniferous lavas, a direct baked contact test with the lavas was not possible.

Paleomagnetic experiments

Data from a total of 101 samples, individually drilled and oriented in the field, are presented

here representing 26 paleomagnetic sites from the Ringerike Group and related rocks. Most samples have been thermally demagnetized using a Schonstedt TSD-1 specimen demagnetizer. It should be noted that temperatures indicated for thermally demagnetized samples are uncalibrated oven temperatures, and may differ (~ 5–10 °C) from the actual internal temperature of the sample. Several representative Ringerike samples were partially demagnetized using a Schonstedt GSD-1 alternating field demagnetizer. For the Ringerike sediments, alternating field demagnetization up to 100 mT had little effect on observed magnetization directions, and reduced NRM intensities by no more than about 25%. Isothermal remanence magnetization (IRM) acquisition experiments were conducted on eight representative samples before and after thermal treatment. Chemical demagnetization was conducted on 10 representative Ringerike samples. After 226 hours of soaking in 8*N* HCl at 20 °C, virtually no change in magnetic intensity or direction was observed and the leaching experiment was discontinued. All magnetic direction data were obtained using a Schonstedt SSM-1 spinner magnetometer at Dartmouth College, with a noise level below ~ 0.3 mA/m.

Results

Permo-Carboniferous lavas

In general, samples taken from the lava sequence exhibit a randomly directed low temperature component, with an unblocking temperature ranging from 100 ° to 500 °C. An internally consistent high temperature component also exists in these rocks and persists to approximately 700 °C. Thermal demagnetization curves for two representative lava samples are shown in Fig. 3. Mean site directions, plotted in in-situ coordinates, cluster tightly about a combined mean of N197 °E declination, − 29 ° inclination (alpha 95% = 18.0, $k = 47.8$, 3 sites, 9 samples) as shown in Fig. 4A. When mean site directions are corrected for attitude of the individual flows (Fig. 4B), the mean direction becomes N204 °E, − 33 ° (alpha 95% = 13.4, $k = 85.7$) which is nearly identical to the mean obtained by Van Everdingen (1960) of N204 °E, − 36.5 ° (484 lava samples) and the "B" component direction described by Storetvedt et al. (1978).

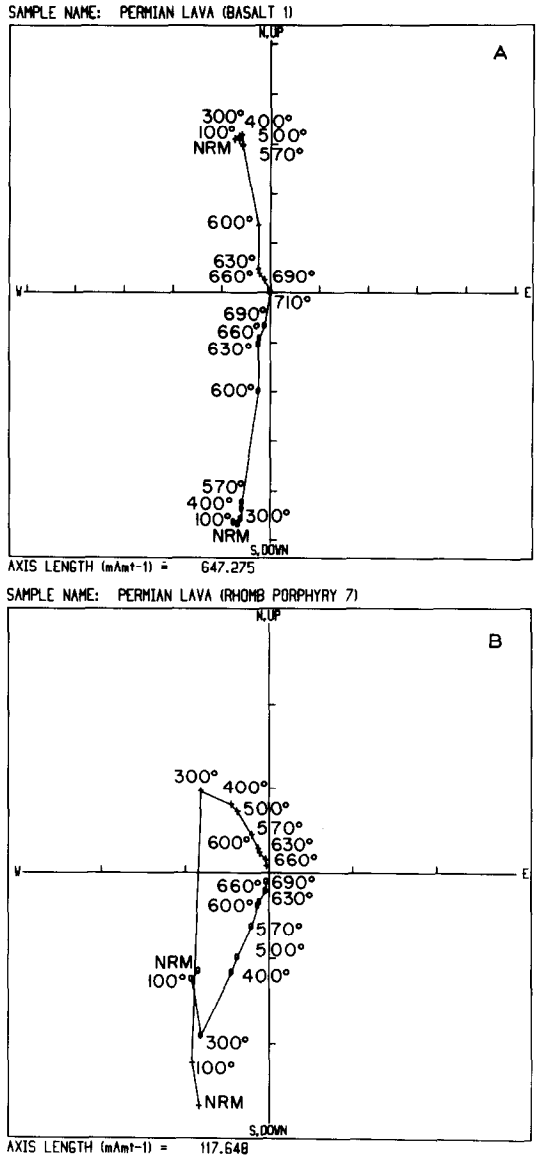


Fig. 3. Zijderveld (orthogonal projection) diagrams showing typical thermal demagnetization behavior of lava samples. Axis ticks are 125 mA/m (A) and 40 mA/m (B). Data are plotted in in-situ coordinates, steps are in °C, circles are vector endpoints projected on the horizontal plane, crosses show projection on the vertical plane perpendicular to the N–S meridian.

Ringerike Sandstone

Two characteristic magnetization components were observed during stepwise thermal demagneti-

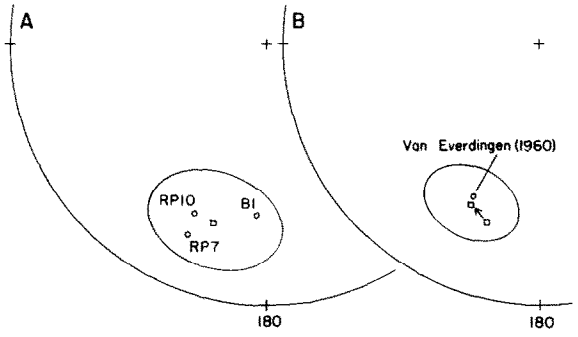


Fig. 4. Equal area stereonet plot of data from Permo-Carboniferous lavas. A. In-situ site means (circles) and combined mean (square with oval of 95% confidence). B, Combined site mean before (square) and after tilt correction (square with oval of 95% confidence) as compared to formation mean (circle) of Van Everdingen (1960). Open symbols indicate points plotted on the upper hemisphere.

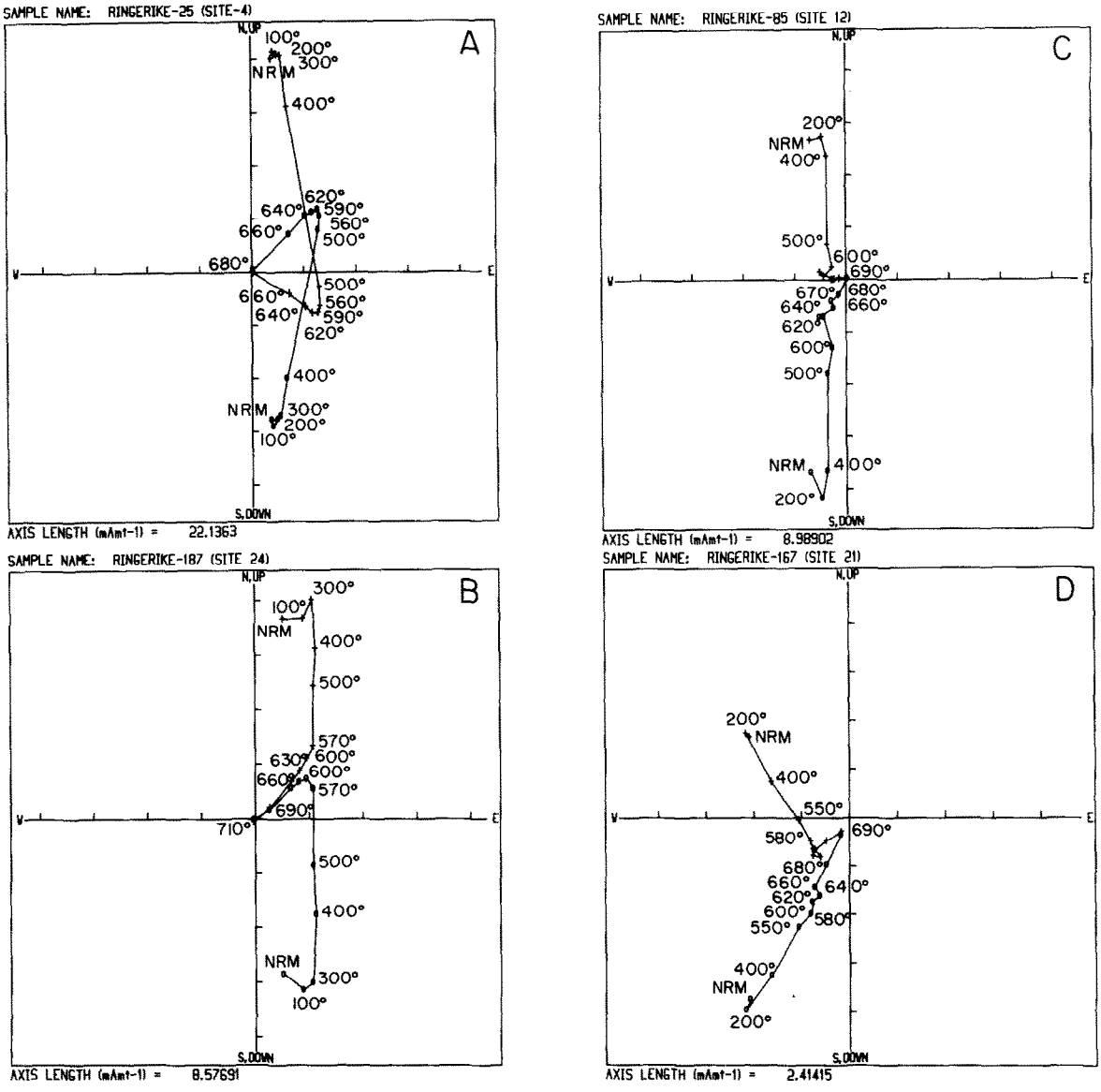


Fig. 5. Zijderveld diagrams showing two different characteristic multicomponent thermal demagnetization curves for the Ringerike Group. All samples show southwest and up directed lower temperature component. A and B show northeast directed higher temperature component. Axis ticks are at 5 mA/m (A) and 2 mA/m (B). C and D show south directed higher temperature component. Axis ticks are at 2 mA/m (C) and 0.5 mA/m (D). All diagrams plotted in in-situ coordinates as in Fig. 3.

TABLE 1

Site mean statistics for low temperature magnetic components observed in the Ringerike Sandstone *

Site	I-decl. (°)	I-incl. (°)	A_{95} (°)	K	R	N	C-decl. (°)	C-incl. (°)	B-decl. (°)	B-incl. (°)
Section A										
15	185.9	−32.9	11.9	108.0	2.98	3	207	−34	220	−40
16-0.15	180.3	−21.4	17.0	53.8	2.96	3	194	−28	215	−43
16-3 [a]	177.4	−26.4	42.5	36.6	1.97	2	196	−34	221	−48
17	181.2	−28.9	4.7	674.6	2.99	3	200	−34	218	−46
18	190.3	−40.1	17.9	48.5	2.96	3	214	−38	196	−47
20	191.8	−43.5	5.4	292.3	3.99	4	207	−36	207	−36
21	193.0	−41.5	21.1	35.1	2.94	3	208	−35	201	−51
22	192.5	−36.5	4.0	950.2	2.99	3	205	−29	205	−29
23	188.2	−37.4	6.1	409.1	2.99	3	200	−30	200	−30
24	190.6	−40.0	9.2	100.6	3.97	4	205	−32	194	−50
I-mean	188.0	−36.0	5.1	101.4	8.92	9	–	–	–	–
C-mean	–	–	3.8	189.1	8.96	9	204	−33	–	–
B-mean	–	–	6.9	57.0	8.86	9	–	–	206	−42
Section B										
3	194.3	−58.3	5.1	573.7	2.99	3	217	−48	176	−10
4	192.3	−50.0	6.4	205.5	3.99	4	211	−41	176	4
5	192.2	−43.2	13.1	89.3	2.98	3	206	−34	189	−9
6	189.5	−46.9	8.4	121.0	3.98	4	207	−39	179	8
7	189.8	−42.9	5.4	519.6	2.99	3	205	−35	183	13
8	193.1	−49.6	5.6	264.7	3.99	4	211	−41	188	−23
9	199.7	−42.0	4.5	417.3	3.99	4	213	−31	198	−17
10	182.4	−56.2	9.4	173.6	2.99	3	207	−49	192	−38
11	193.1	−44.6	6.9	318.8	2.99	3	208	−36	198	−20
12	188.2	−43.2	4.9	245.7	4.98	5	204	−36	204	−28
I-mean	192.0	−48.0	3.9	156.3	9.94	10	–	–	–	–
C-mean	–	–	3.9	149.0	9.94	10	209	−39	–	–
B-mean	–	–	11.6	18.3	9.51	10	–	–	188	−12

Low temperature results

	A_{95} (°)	K	R	N	Decl. (°)	Inc. (°)
In-situ (I)	3.9	74.3	18.8	19	190	−42
Tilt corrected (C)	2.95	130.1	18.9	19	207	−36
Tilt corrected (B)	9.96	12.3	17.5	19	196	−27

* "I" indicates in-situ directions; "C" indicates directions corrected for Carboniferous tilt; "B" indicates directions after resultant bedding attitudes are rotated to horizontal; [a] indicates sites which are not statistically significant. Note: Site 16-3 is not included in mean calculations.

zation (Fig. 5). In both cases, a magnetically stable but slightly lower temperature component, with consistent southwesterly and steeply up directions, was observed in all but two sample sites (Fig. 5). A higher temperature component, having either a northeasterly or southwesterly declination, and a relatively shallow positive or negative inclination, was also observed in all but one site. The component of lower magnetic stability generally has unblocking temperatures of up to 600 °C or more.

TABLE 2

Site mean statistics for high temperature magnetic components observed in the Ringerike Sandstone *

Site	I-decl. (°)	I-incl. (°)	A_{95}	K	R	N	B-decl. (°)	B-incl. (°)
Section A								
15	50.3	−50.2	6.9	95.9	5.95	6	2	−40
16-0 [b]	169.6	−6.5	3.4	507.1	4.99	5	187	−40
16-0.15	22.2	−28.1	13.8	81.4	2.98	3	2	−1
16-3	34.0	−25.5	10.2	56.7	4.93	5	8	−8
17	50.6	−44.9	11.6	34.4	5.85	6	4	−35
18	62.0	−33.9	29.5	10.7	3.72	4	54	−34
20	208.9	−0.4	15.8	34.8	3.91	4	208	10
21	197.3	20.1	15.2	66.3	2.97	3	192	11
22	195.2	−1.0	11.6	113.6	2.98	3	194	6
23	212.7	3.1	5.6	492.0	2.99	3	200	11
24	54.3	−41.7	14.2	42.6	3.93	4	44	−37
I-mean	35.0	−25.5	14.7	11.7	9.23	10	–	–
B-mean	–	–	13.6	13.5	9.34	10	20	−20
Section B								
1	196.8	−38.9	20.7	36.3	2.94	3	198	14
2	24.3	34.5	6.9	124.8	4.97	5	27	−18
3	43.2	1.3	10.6	53.1	4.92	5	37	−6
4	48.9	21.5	4.6	213.5	5.97	6	49	0
5	30.9	10.4	5.4	157.1	5.97	6	38	−12
6	26.2	17.3	30.6	17.3	2.88	3	40	−24
8 [a]	38.1	−1.0	74.3	3.8	2.47	3	47	−22
9 [a]	47.8	−24.5	17.8	199.3	1.99	2	63	−42
10	42.3	13.7	10.4	142.3	2.99	3	52	−12
11	177.9	−28.3	33.0	15.0	2.87	3	186	−5
12	218.5	−14.5	19.4	23.4	3.87	4	212	26
I-mean	30.7	20.6	12.2	18.7	8.57	9	–	–
B-mean	–	–	11.4	21.3	8.62	9	33	−12

High temperature results

	A_{95}	K	R	N	Decl. (°)	Incl. (°)
Normal sites (I)	18.7	6.32	10.3	12	39.9	−11.3
Reversed sites (I)	18.5	11.63	6.48	7	201.6	−8.7
All sites (I)	13.9	6.85	16.4	19	32.9	−3.3
Normal sites (B)	13.2	11.74	11.1	12	30	−20
Reversed sites (B)	9.6	40.5	6.85	7	198	11
All sites (B)	9.1	14.7	17.8	19	26	−16

* "I" indicates in-situ directions; "B" indicates directions after 2 stage tilt correction; [a] indicates sites which are not statistically significant; [b] indicates Permo-Carboniferous dike. Note: sites 16-0, 8 and 9 are not included in mean calculations.

Some samples did not exhibit this lower temperature component (e.g. sites 1 and 2), while in several samples this was the only component successfully isolated (e.g. site 7). High temperature components were generally defined by demagnetization steps between 620° and 700°C (Fig. 5). De-

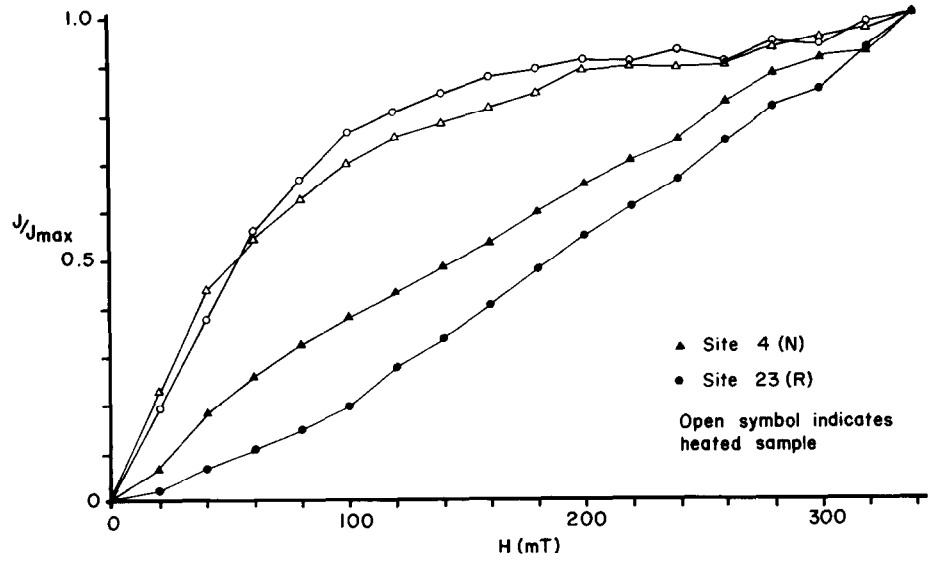
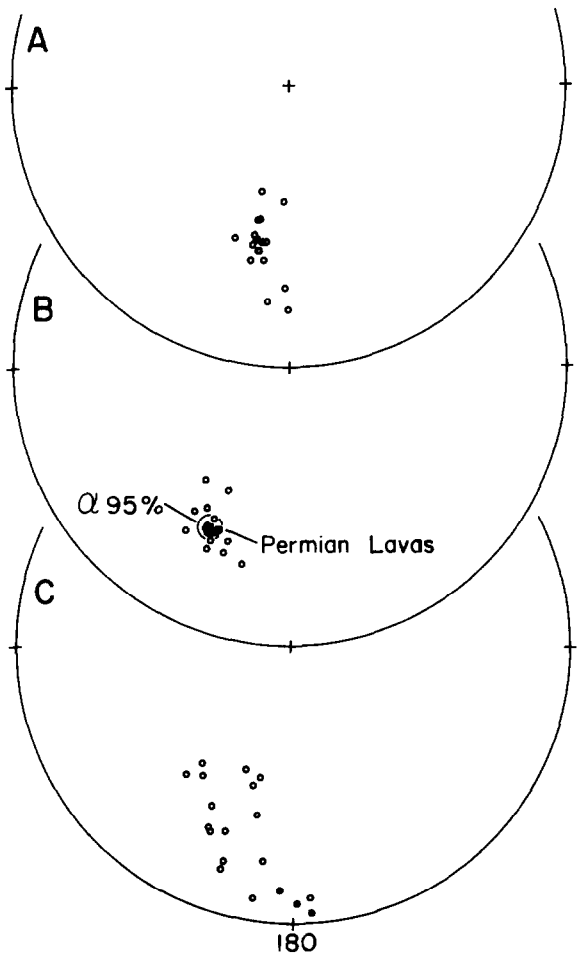


Fig. 6. Plot of isothermal remanent magnetization (IRM) acquisition (J/J_{max}) with increasing applied field (H) for representative Ringerike samples.



magnetization was discontinued when sample intensities approached the noise level of the instrument. Component directions for each site were determined graphically from Zijderveld (orthogonal projection) diagrams and calculated using a principal component analysis method similar to that described by Kirschvink (1980). In-situ and tilt-corrected site mean directions for lower and higher stability components are given in Tables 1 and 2 respectively.

Isothermal remanance magnetization (IRM) acquisition curves for multicomponent samples with both northerly and southerly directed high temperature components exhibit a nearly linear increase in acquired remanence to a maximum applied field of 320 mT (Fig. 6). Duplicate specimens of these samples which had been heated to ~ 700 °C during thermal demagnetization showed an increase in IRM acquisition at lower applied field strengths (Fig. 6). These results suggest an addition or change in the magnetic

Fig. 7. Equal area plot of lower temperature component Ringerike site means. A. Data plotted in in-situ coordinates. B. Data plotted in post-Permian tilt corrected coordinates including 95% circle of confidence of combined mean, and Permo-Carboniferous lava mean of Van Everdingen (1960). C. Data plotted after 2 stage tilt correction. Open (closed) symbols indicate points plotted on the upper (lower) hemisphere.

mineralogy of the samples during thermal demagnetization, possibly by reduction of hematite to magnetite (Chamalaun and Creer, 1964). This may explain why a high temperature component could not be successfully isolated in a few cases.

Paleomagnetic fold test

Because there is evidence for at least two stages of deformation in the Oslo Graben, in-situ magnetic directions for Ringerike rocks require a two step tilt correction: Step (1)-Bedding attitudes and mean magnetic directions for each site were corrected for bedding tilt of the Carboniferous sediments which unconformably overly Ringerike rocks. Step (2)-Mean magnetic directions for each site were then corrected by rotating the bedding attitudes resulting from Step 1 to horizontal.

When lower temperature component directions for Ringerike sediments (Fig. 7A, Table 1) are corrected for Carboniferous bedding tilt, they yield an overall mean direction of N207° E, – 36° which is nearly identical to the mean Permo-Carboniferous lava direction (Fig. 7B). Alternatively, if these sites are given the full bedding correction (steps 1 and 2 above), mean site directions become dispersed along a great circle (Fig. 7C). Using the criteria of McElhinny (1964), the lower temperature magnetization component passes a negative fold test (i.e. post folding) at the 99% confidence level. This indicates that the lower stability component is clearly a secondary magnetization which post-dates the major pre-Carboniferous (Caledonian) folding of the Ringerike sediments.

Higher temperature, in-situ site mean directions from the Ringerike Group are relatively dispersed on both upper and lower hemispheres (Fig. 8A). When the two-fold tilt correction described above is applied to the high temperature mean directions, they pass the fold test at a greater than 95% confidence level. Therefore, this higher temperature direction represents a pre-folding component of magnetization (Fig. 8B). Because sites 15–18 do not lie directly under the Permo-Carboniferous lavas, and because there are one or more faults separating these sites from others in the section (Fig. 2A), the post-Carboniferous tilt correction of step 1 may not be appropriate for these sites. Therefore, in order to better estimate the true high temperature means for sites 15–18, step 1 of the tilt correction was modified so that the mean overprint direction would coincide with the cor-

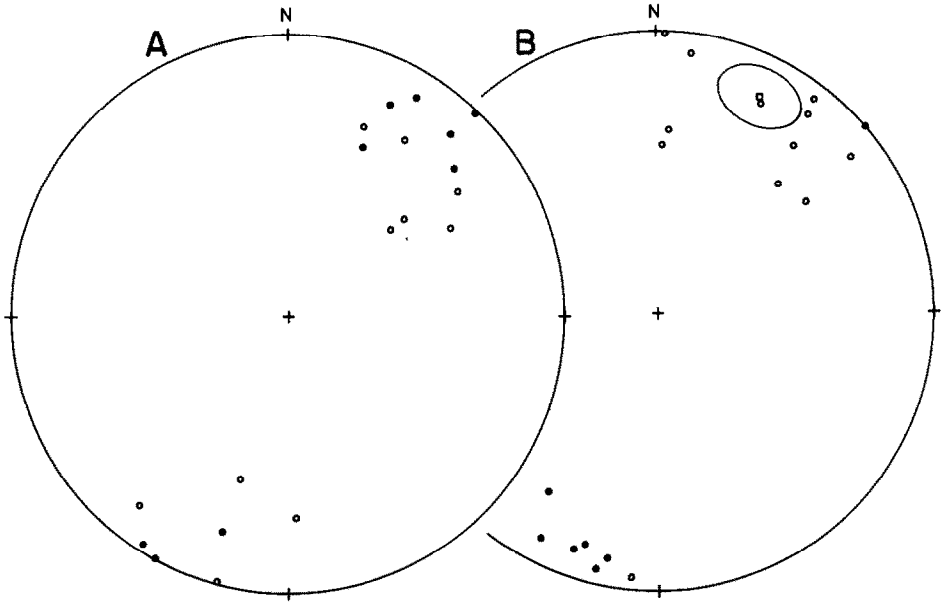


Fig. 8. Equal area plot of Ringerike higher temperature component site means. A. Data plotted in in-situ coordinates. B. Data plotted after 2 stage tilt correction. Combined mean for all sites (square) is shown with oval of 95% confidence. Open (closed) symbols indicate points plotted on the upper (lower) hemisphere.

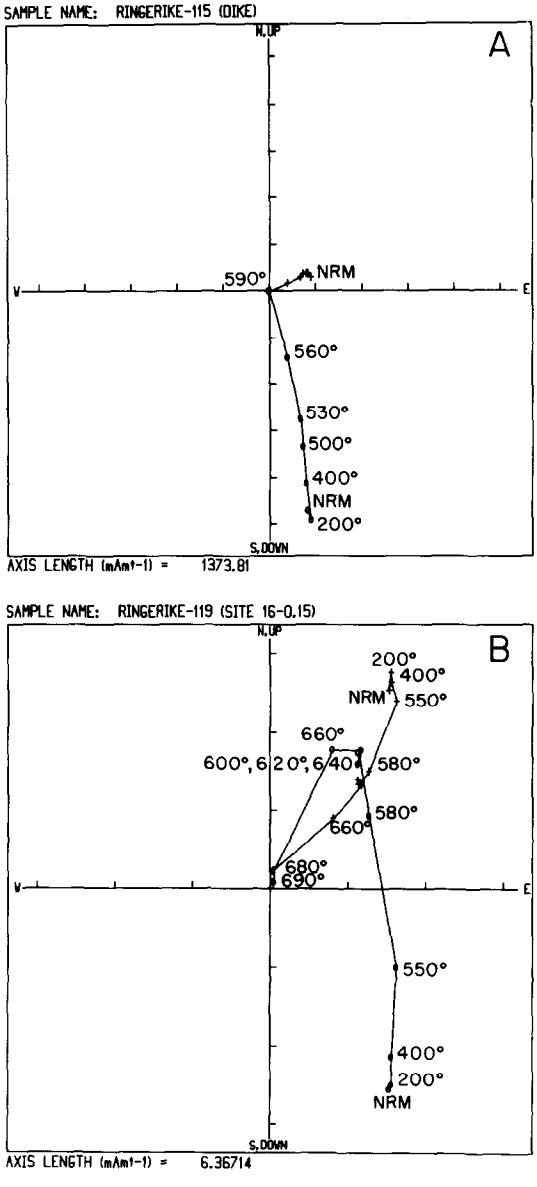


Fig. 9. Zijderveld diagram of thermal demagnetization behavior. A. Diabase dike unit intruding basal Ringerike sediments at Section A. Axis ticks are at 250 mA/m. B. Sample taken 0.15 m from diabase dike near the base of the Ringerike section at Kroksund showing low temperature, south directed component, and high temperature, northeast directed characteristic direction. Axis ticks are at 2 mA/m. Both diagrams plotted in in-situ coordinates as in Fig. 3.

rected overprint direction of all other sites. This had the effect of increasing the step 1 tilt correction approximately 10°, and changing the strike of the plane of rotation slightly. The high temperature directions were then tilt corrected as described in step 2 (above). It should be noted, however, that either tilt correction method produced a positive fold test at a greater than 95% confidence level.

Baked contact test

Because Permo-Carboniferous lavas overlie basal Carboniferous sediments, Ringerike rocks could not be sampled in direct contact with the lavas. However, the coincidence of lower temperature overprint directions with the mean direction for the lavas (Fig. 7B) strongly suggests that the overprint was acquired in the sediments at the time of lava eruption.

At site 16 on the west shore of Kroksund (Fig. 2A), an approximately 1.2 m diabase dike of probable Permo-Carboniferous age was sampled. A high temperature component of magnetization was isolated for this dike unit yielding a mean direction of N170°, −7° (site 16-0, Table 2). A representative thermal demagnetization curve for the dike rock is shown in Fig. 9A. Adjacent to both sides of this dike is an approximately 0.2 m thick alteration zone. Ringerike sediments were sampled at distances of 0.15 m (site 16.15, Table 2) and 3.0 m (site 16-3, Table 2) from the dike contact. In both cases, a high temperature component similar to the mean Ringerike direction and radically different from the dike was isolated in the baked sediments. An example of the thermal demagnetization behavior of a baked sample is shown in Fig. 9B.

The removal of a lower stability magnetic direction, which can be attributed to the dike, indicates a pre-intrusion age for the higher temperature component recognized in the baked sediments. It is unlikely that both the dike unit and the sediment were completely overprinted in the Permo-Carboniferous, because the mean dike direction (decl. = 169.6°, incl. = −6.5°, Table 2) differs significantly from the mean Permo-Carboniferous lava direction (decl. = 204°, incl. = −36.5°), even when corrected for possible post-Permian tilt (decl. = 178°, incl. = −22°).

It is interesting to note that the mean dike direction, when tilt corrected for the adjacent Ringerike sediments, yields a direction closer to

the expected Permo-Carboniferous mean (decl. = 187°, incl. = −40°, Table 2). Because the dike does not carry a time-averaged paleomagnetic direction, this result may simply be coincidence. For example, an anomalous magnetic direction captured by the dike at the time of intrusion could, upon tilt correction, fortuitously correspond to the mean Permo-Carboniferous direction. Alternatively, the necessity of a tilt correction may indicate that the dike is pre-folding. This would require a previously unrecognized intrusive event in Early Devonian time to account for the dike, or rapid folding of the sequence in Late Carboniferous time, after intrusion of the dike but prior to eruption of the main lava sequence. Further constraints on the age of dike intrusion, and paleomagnetic directions from additional dikes, are necessary before these possibilities can be resolved.

Interpretation and conclusions

Magnetic mineralogy and NRM components

Several lines of evidence suggest that both the higher and lower stability NRM components in the Ringerike sandstone are carried by hematite: (1) Unblocking temperatures for the lower temperature component are usually above the Curie temperature for magnetite or titanomagnetites, while the maximum unblocking temperature for the higher temperature component is generally around 690°C. (2) IRM acquisition curves demonstrate that samples do not become saturated in applied fields up to 320 mT. (3) Petrographic studies indicate that detrital magnetite grains are virtually absent in Ringerike sandstones while detrital (specular) hematite is common (Turner, 1974c, p. 224).

The high-temperature component observed in Ringerike rocks is interpreted as primary based on the occurrence of stratigraphically correlative, antipodal dual polarities (i.e. overlapping circles of 95% confidence for normal and reversely magnetized sites), a positive fold test, baked contact test results, and consistent magnetic behavior in sedimentologically variable samples.

Detrital hematite, hematite formed by in-situ alteration, and pigment hematite are all recognized in Ringerike rocks by Turner (1974c). It is likely that the primary magnetic component observed is a detrital remanent magnetization (DRM) carried largely by specular hematite. It is also possible that this primary direction is carried, in part, by a chemical remanent magnetism (CRM) component related to alteration or pigment hematite. If a significant CRM component exists, it was probably acquired on a timescale comparable to sedimentological events observed in the sequence (i.e. the fluvial cycle) because: (1) red (pigmented) and non-red lithologies are distinctly interstratified; (2) there is an overall bias of the redder lithologies towards the finer grained (e.g. overbank) facies and; (3) no difference was observed in magnetic behavior between red and non-red paleomagnetic samples. This CRM may have originated from rapid oxidation of hydrous iron oxide minerals shortly after deposition in the fluvial system (Turner, 1974c, 1979). The sedimentological bias of pigmented lithologies towards the finer-grained facies may be due to variations in oxidation–reduction conditions within the fluvial environment such as described by Friend (1966).

The amount of time represented by Ringerike rocks can be estimated by considering known sedimentation rates for analogous rocks. The Ringerike sequence studied here encompasses approximately 1200 m of intercalated fluvial and marine shoreline facies (Turner, 1974a, b). Mean accumulation rates for foreland basin sediments range from 10^{-2} to 10^{0} m/1000 years (Schwab, 1976). Based on a maximum sedimentation rate of 1 m/1000 years, the Ringerike Group has a probable time span of at least 1.2×10^{6} years. This amount of time is more than sufficient to average out geomagnetic secular variation and provide a time-averaged primary paleomagnetic direction.

The lower temperature magnetic component of the Ringerike Sandstone is interpreted as a viscous partial thermal remanent magnetization (VPTRM), possibly in combination with a minor CRM, acquired during Permo-Carboniferous time. Unblocking temperatures for this component commonly extend to 620°C, which is noticeably high compared to a maximum paleotemperature of

300 °C (discussed above). Igneous activity within the Oslo Graben is believed to have lasted approximately 50 million years (Oftedahl, 1978; Ramberg and Spjeldnaes, 1978). When these conditions are applied to the time–temperature curves of Pullaiah et al. (1975) they predict unblocking temperatures of approximately 500 °C, which is ~ 120 °C lower than observed. However, the time–temperature curves of Middleton and Schmidt (1982), derived from Walton (1980), indicate unblocking temperatures of ~ 600 °C for these same geologic conditions. Therefore, results from the Ringerike Sandstone are in better agreement with the time–temperature model of Walton (1980).

There is petrologic evidence indicating that some degree of alteration of iron oxide and iron-rich phyllosilicate minerals occurred after deposition (Turner, 1974c). If alteration occurred over a relatively short period of time, it may have resulted in secondary CRM with a consistent magnetic direction. If this is the case, then any CRM overprint must have occurred during the Permo-Carboniferous (Kiaman reversed polarity interval) because of the consistent polarity and direction observed.

In rocks with both pigmentary hematite (from alteration) and specular hematite, the pigment nearly always possesses a magnetization which is less stable during thermal demagnetization than the specular hematite (Collinson, 1974). An alternative explanation for the lower temperature component observed in Ringerike rocks is that a VPTRM reset any pre-existing CRM (i.e. pigment) components while not completely affecting the primary DRM (specularite) component direction.

Tectonic implications

The characteristic component directions for the Ringerike rocks studied here yield a paleopole position of 19 °N latitude, 164 °E longitude ($dp = 4.8$, $dm = 9.3$, Fig. 10). This paleopole is very similar to the direction obtained previously from the Ringerike sandstone by Storetvedt et al. (1967) of 21 °N, 159 °E. Because the Ringerike direction is pre-folding, the timing of magnetization is distinctly different than the syn-tectonic magnetization recently proposed for the Kvamshesten Basin of western Norway (Torsvik et al., 1986). The Ringerike direction is interpreted as a primary, Late Silurian–Early Devonian (Ludlow to Downton) time-integrated paleopole position.

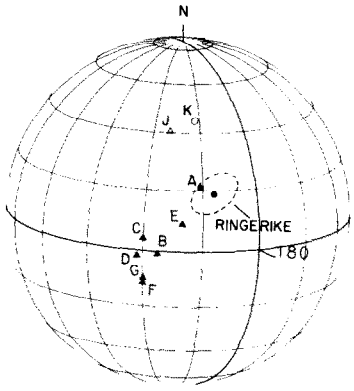


Fig. 10. Paleopole position for the Ringerike Sandstone shown with oval of 95% confidence, along with paleopole data from other studies of similar age rocks in Norway (circles) and the British Isles (triangles). A. Storetvedt et al. (1967), Ringerike Sandstone, uS-1D Norway. B. McMurray (1970), Lower Old Red Sandstone, uS-1D Midland Scotland. C. Sallomy and Piper (1973), lower Old Red Sandstone, uS-1D Midland Scotland. D. Torsvik (1985), lower Old Red Sandstone, uS-1D Midland Scotland. E. Storhaug and Storetvedt (1985), lower Old Red Sandstone, uS-1D Northern Scotland. F. Thorning (1974), Cheviot Volcanics, 1D Northern England. G. Chamalaun and Creer (1964), lower Old Red Sandstone, 1D Welsh Borderlands. Permian–Carboniferous paleopole positions (open symbols) are shown for Britain (J: Turner and Tarling, 1975) and Norway (K: Van Everdingen, 1960; this paper).

Extensive reviews of British paleomagnetic results can be found elsewhere (e.g. Briden et al., 1973, 1984; Piper, 1979; Tarling, 1985; and others), therefore only a few representative paleopole results will be discussed here for the purpose of comparison (Fig. 10). Probably the most well studied Siluro-Devonian age rocks of the British Isles are the Lower Old Red Sandstone lavas and sediments of the Stonehaven, Dunnottar, Crawton and Arbuthnott Groups occurring in the Strathmore Region. Fauna from the lowest unit studied paleomagnetically (Stonehaven Group) indicates a Downtonian age (Armstrong and Paterson, 1970) and have been directly compared to the faunal assemblage of the Ringerike Sandstone (Heintz,

1969; Turner and Turner, 1974). Lavas occurring approximately 3000 m above the base of the section (Crawton Group) yield chronometric ages of between 405–410 Ma (Thirlwall and Astin, 1983). Therefore Lower Old Red Sandstone rocks of the Midland Valley are very similar in age to the Ringerike sediments.

A paleopole direction for the Midland Valley Lavas is given by McMurray (1970) which includes both normal and reversed polarity directions and is constrained by a paleomagnetic fold test (Fig. 10, *B*). A second study (Sallomy and Piper, 1973), also including dual polarity directions, a positive fold test, baked contact test and conglomerate test provides similar results (Fig. 10, *C*). Most recently, these rocks were re-examined (Torsvik, 1985) and again the paleomagnetic directions obtained support the earlier results (Fig. 10, *D*). The paleomagnetic fold tests, and other field tests for magnetization age, constrain these magnetic directions to pre-middle Devonian. These studies all indicate a near equatorial paleopole for the Midland Valley Region in Late Silurian–Early Devonian time.

According to Tarling (1985), there are as yet no precise Late Silurian–Early Devonian paleomagnetic data for Old Red Sandstone rocks of northern Scotland and there is currently no evidence to suggest a paleopole which is radically different than the Midland Valley region. However, a study of the Lower Old Red (Sarclet) Sandstone of the Orcadian Basin (Storhaug and Storetvedt, 1985) suggest a paleomagnetic pole position which is intermediate between the Ringerike pole and similar age rocks from the Midland Valley (Fig. 10, *E*). Dual polarity magnetic directions were obtained from a high-temperature component observed in these rocks which is interpreted as a primary DRM. However, sampling conditions were such as to preclude any paleomagnetic field test to constrain the age of the magnetization components observed.

Early Devonian paleopole positions from sediments and lavas elsewhere in the British Isles are also generally consistent with results from the Midland Valley Region. For example, results from the Cheviot Hills of northern England, dated by K/Ar methods as Early Devonian (Thorning, 1974), plot reasonably close to the Midland Valley paleopoles (Fig. 10, *F*). A study of Old Red Sandstone sediment of the Welsh Borderlands (Chamalaun and Creer, 1964) also indicate a similar paleopole position (Fig. 10, *G*) and is supported by a paleomagnetic fold test.

An approximately 20° latitudinal separation between the British terranes and Baltica in Late Silurian–Early Devonian time is indicated by comparison of well constrained, similar age paleomagnetic pole positions. This amount of separation was minimized by the late Carboniferous because Permo-Carboniferous paleopole positions from Britain (Turner and Tarling, 1975) and Norway (Van Everdingen, 1960; Storetvedt et al., 1978; this paper) are similar (within errors). The similarity between paleopoles of probable Late Devonian–Early Carboniferous age from the Kvamshesten Basin of western Norway (Torsvik et al., 1986), and Middle to Late Devonian age Midland Valley poles (e.g. Briden et al., 1973) may indicate possible closure between the two areas as early as Middle Devonian time. This possibility cannot be resolved however, until primary, field tested Middle Devonian age magnetization directions can be established for rocks of the Baltic craton.

Acknowledgements

I would like to thank Bjørn Larsen for kind direction and hospitality in the field. I also gratefully acknowledge D.V. Kent, T.H. Torsvik and two anonymous reviewers for their comments and critical review of this manuscript. This research was supported in part by a grant from the R.E. Stoiber Field Fund, Department of Earth Sciences, Dartmouth College, and from the O.C. Marsh Fund, National Academy of Sciences, administered by Sigma Xi. This paper is dedicated to the memory of Noye M. Johnson.

References

Aldridge, R.J., 1984. Thermal metamorphism of the Silurian strata of the Oslo Region, assessed by conodont color. Geol. Mag., 4: 347–349.

Armstrong, M. and Paterson, I.B., 1970. The Lower Old Red Sandstone of the Strathmore Region. Inst. Geol. Sci., Her

Majesty's Stationary Office, London, Rep. No. 70/12, 24 pp.

Bergstrøm, S.M., 1980. Conodonts as paleotemperature tools in Ordovician rocks of the Caledonides and adjacent areas in Scandinavia and the British Isles. Geol. Føren. Stockholm Førh., 102 (4): 377–392.

Briden, J.C., Morris, W.A. and Piper, J.D.A., 1973. Paleomagnetic studies in the British Caledonides, VI. Regional and global implications. Geophys. J.R. Astron. Soc., 34: 107–134.

Briden, J.C., Turnell, H.B. and Watts, D.R., 1984. British paleomagnetism, Iapetus Ocean and the Great Glen Fault. Geology, 12(7): 428–431.

Bullard, E.C., Everett, J.E. and Smith, A.G., 1965. Fit of the continents around the Atlantic: a symposium on continental drift, IV. Philos. Trans. R. Soc. London, Ser. A, 258: 41–51.

Chamalaun, F.H. and Creer, K.M., 1964. Thermal demagnetization studies on the Old Red Sandstone of the Anglo-Welsh Cuvette. J. Geophys. Res., 69(8): 1607–1616.

Collinson, D.W., 1974. The role of pigment and specularite in the remanent magnetism of red sandstones. Geophys. J.R. Astron. Soc., 38: 253–264.

Creer, K.M., 1968. Palaeozoic palaeomagnetism. Nature, 219: 246–250.

Friend, P.F., 1966. Clay fractions and colours of some Devonian red beds in the Catskill Mountains, U.S.A. J. Geol. Soc. London, 122: 273–292.

Hagemann, F., 1966. Silurian bentonites in the Oslo Region. Nor. Geol. Unders., 242: 44–61.

Heintz, A., 1969. New agnaths from Ringerike sandstone. Skr. Nor. Vidensk.-Akad., I, N.S., 26: 1–28.

Henningsmoen, G., 1978. Sedimentary rocks associated with the Oslo Region lavas. In: J.A. Dons and B.T. Larsen (Editors), The Oslo Paleorift—a Review and Guide to Excursions. I.U.C.G., Sci. Rep., 46: 17–24.

Hoffman, J. and Hower, J., 1979. Clay mineral assemblages as low grade metamorphic geothermometers: application to the thrust faulted disturbed belt of Montana. Soc. Econ. Paleontol. Mineral. Spec. Publ., 26: 55–79.

Holtedahl, O., 1953. Norges Geology. Nor. Geol. Unders., 164: 1118 pp.

Kent, D.V. and Keppie, J.D., in-press. Silurian–Permian paleo-continental reconstructions and circum-Atlantic tectonics. In: A.L. Harris and D.J. Fetts (Editors), The Caledonian–Appalachian Orogen. Geol. Soc. London, Spec. Publ., Chapt. 9.

Kirschvink, J.L., 1980, The least-squares line and plane and the analysis of paleomagnetic data. Geophys. J.R. Astron. Soc., 62: 699–718.

Larsen, B.T., 1975. Oslo Region tectonic map. Norwegian Geotechnical Institute, Proj. no. 74628.

Larsen, B.T., 1978. Krokskogen lava area. In: J.A. Dons and B.T. Larsen (Editors), The Oslo Paleorift—a Review and Guide to Excursions. Nor. Geol Unders.–I.U.C.G., Sci. Rep., 46: 151–162.

Lie, L.G., Storetvedt, K.M. and Gjellestad, G., 1969. The palaeomagnetism of the Kvamshesten Old Red Sandstone sequence, southwest Norway. Nor. Geol. Tidsskr., 49: 241–249.

McElhinny, M.W., 1964. Statistical significance of the fold test in paleomagnetism. Geophys. J.R. Astron. Soc., 8: 338–340.

McMurray, E.W., 1970. Palaeomagnetic results from Scottish lavas of lower Devonian age. In: S.K. Runcorn (Editor), Palaeogeophysics. Academic Press, London, pp. 253–262.

Middleton, M.F. and Schmidt, P.W., 1982. Paleothermometry of the Sydney Basin. J. Geophys. Res., 87 (B7): 5351–5359.

Morley, C.K., 1986. A classification of thrust fronts. Am. Assoc. Pet. Geol. Bull., 70: 12–25.

Oftedahl, C., 1978. Main geologic features of the Oslo Graben. In: J.B. Ramberg and E.R. Neumann (Editors), Tectonics and Geophysics of Continental Rifts. Reidel, Dordrecht, pp. 149–165.

Olaussen, S., 1981. Marine incursions in the upper Paleozoic sedimentary rocks of the Oslo Region, southern Norway. Geol. Mag., 118: 281–288.

Palmer, A.R., 1983. The decade of North American geology 1983 geologic timescale. Geology, 11: 503–504.

Piper, J.D.A., 1979. Aspects of Caledonian paleomagnetism and their tectonic implications. Earth Planet. Sci. Lett., 44: 176–192.

Pullaiah, G., Irving, E., Buchan, K.L. and Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift. Earth Planet. Sci. Lett., 28: 133–143.

Ramberg, I.B., 1976. Gravity interpretation of the Oslo Graben and associated igneous rocks. Nor. Geol. Unders., 325: 1–194.

Ramberg, I.B. and Spjeldnaes, N., 1978. The tectonic history of the Oslo Region. In: J.B. Ramberg and E.R. Neumann (Editors), Tectonics and Geophysics of Continental Rifts. Reidel, Dordrecht, pp. 167–194.

Sallomy, J.T. and Piper, J.D.A., 1973. Paleomagnetic studies in the British Caledonides, IV. Lower Devonian lavas of the Strathmore Region, Scotland. Geophys. J.R. Astron. Soc., 34: 47–68.

Schwab, F.L., 1976. Modern and ancient sedimentary basins: comparative accumulation rates. Geology, 4: 723–727.

Storetvedt, K.M., 1970. The Devonian Paleomagnetic field for Europe. In: S.K. Runcorn (Editor), Palaeogeophysics. Academic Press, London, pp. 247–252.

Storetvedt, K.M. and Gjellestad, G., 1966. Paleomagnetic investigation of an Old Red Sandstone formation, southern Norway. Nature, 212: 59–61.

Storetvedt, K.M., Halvorsen, E. and Gjellestad, G., 1967. Thermal analysis of the natural remanent magnetism of some upper Silurian red sandstones in the Oslo Region. Tectonophysics, 5(5): 413–426.

Storetvedt, K.M., Pederson, S., Løvlie, R. and Halvorsen, E., 1978. Paleomagnetism in the Oslo Rift. In: J.B. Ramberg and E.R. Neumann (Editors), Tectonics and Geophysics of Continental Rifts. Reidel, Dordrecht, pp. 289–296.

Storhaug, K. and Storetvedt, K.M., 1985. Paleomagnetism of the Sarclet sandstone (Orcadian Basin): age perspectives. Scott. J. Geol., 21(3): 275–284.

Sundvoll, B., 1978. Rb/Sr relationship in the Oslo igneous rocks. In: J.B. Ramberg (Editor), Petrology and Geochemistry of Continental Rift Basins. Reidel, Dordrecht, pp. 181–184.

Tarling, D.H., 1985. Paleomagnetic studies of the Orcadian Basin. Scott. J. Geol., 21(3): 261–273.

Thirlwall, M.F. and Astin, T.R., 1983. Discussion on implications for Caledonian plate tectonic models of chemical data from volcanic rocks of the British Old Red Sandstone. J. Geol. Soc. London, 138: 123–138.

Thorning, L., 1974. Paleomagnetic results from lower Devonian rocks of the Cheviot Hills, northern England. Geophys. J.R. Astron. Soc., 36: 487–496.

Torsvik, T.H., 1985. Magnetic properties of the lower Old Red Sandstone lavas in the Midland Valley, Scotland; paleomagnetic and tectonic considerations. Phys. Earth Planet. Inter., 39: 194–207.

Torsvik, T.H., Sturt, B.A., Ramsay, D.M., Kisch, H.J. and Bering, D., 1986. The tectonic implications of Solundian (Upper Devonian) magnetization of the Devonian rocks of Kvamshesten, western Norway. Earth Planet. Sci. Lett., 80: 337–347.

Turner, P., 1974a. Lithostratigraphy and facies analysis of the Ringerike Group, Oslo Region. Nor. Geol. Unders., 314: 101–131.

Turner, P., 1974b. Marine calcarenites from the Ringerike Group (stage 10) of southern Norway. Nor. Geol. Tidsskr., 54: 1–12.

Turner, P., 1974c. Origin of red beds in the Ringerike Group (Silurian) of Norway. Sediment. Geol., 12: 215–235.

Turner, P., 1979. The paleomagnetic evolution of continental red beds. Geol. Mag., 116(4): 289–301.

Turner, P. and Tarling, D.H., 1975. Implications of new paleomagnetic results from the Carboniferous System of Britain. J. Geol. Soc. London, 131: 469–488.

Turner, P. and Turner, S., 1974. Thelodonts from the upper Silurian of Ringerike, Norway. Nor. Geol. Tidsskr., 54: 183–192.

Van Everdingen, R.O., 1960. Palaeomagnetic analysis of Permian extrusives in the Oslo Region, Norway. Studies on the Igneous Rock Complex of the Oslo Region, XVII. Skr. Nor. Vidensk.-Akad. Oslo, 1: 1–80.

Walton, D., 1980. Time–temperature relations in the magnetization of assemblies of single domain grains. Nature, 280(7): 245–247.

Worsley, D., Aarhus, N., Bassett, M.G., Howe, M.P.A., Mørk, A. and Olaussen, S., 1983. The Silurian succession of the Oslo Region. Nor. Geol. Unders., 72(384): 57 pp.